BrewBot: A Physiological-State-Aware Drink Server

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PLACEHOLDER — Fig. 1 (teaser, full width)

Photo: the arm handing a filled glass to a seated person,
or the arm mid-mime with the user's hand raised in frame.
Should read as interaction, not as hardware.

Figure 1: BrewBot offering and serving a drink in the lab kitchen. The arm hangs from a ceiling rail, so it can travel between the sink, the coffee machine and the person it is serving.

Abstract

Wearable sensors make latent human states such as thermal discomfort, arousal and fatigue continuously legible, which lets a service robot act before it is asked. Acting first, however, moves the difficulty from detection to reply: the robot speaks at a moment the person did not choose, and that person may be typing, holding something, or simply unwilling to talk. We present BrewBot, a ceiling-mounted 6-DOF arm on a linear rail in a shared lab kitchen that infers a user's state from a heart-rate sensor and an Empatica E4 wristband, offers a fitting drink unprompted, and fetches it. The reply may arrive as speech classified by a local language model, as a hand gesture, or as an open palm that waves the conversation away entirely — upon which the arm mimes the menu with its own body and reads a thumbs-up. From building the system we extract three transferable design patterns: make modalities substitutable, separate *what* to offer from *when* to interrupt, and let every non-essential subsystem fail open. We also report where wrist-based physiological sensing fell short of what the interaction needed.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Computing methodologies** → *Robotic planning*.

Keywords

human-robot interaction, proactive robots, physiological sensing, multimodal interaction, gesture interaction, service robotics, ROS 2

1 Introduction

A person working in a shared lab kitchen is continuously in some physical state — warm, tense, tired, restless — and almost never says so out loud. These states are *latent*: they are real, they change behaviour, and they are legible to instruments long before they are legible to a conversation partner. Consumer-grade wearables now stream heart rate, electrodermal activity and skin temperature continuously, which makes it technically straightforward for a robot to notice that someone is overheating and to act on that observation without being asked. The appeal is obvious in a service setting: instead of waiting for a command, the robot offers what the person is about to want.

Acting first, however, moves the hard part of the interaction. Detection is a threshold problem and can be tuned. The difficulty is that a proactive robot speaks at a moment its user did not choose. Whoever it addresses may be typing with both hands, carrying something, mid-sentence with a colleague, wearing headphones, or simply not in the mood to have a conversation with a machine about their heart rate. A command-driven robot never faces this, because the user only ever initiates when they are free to. A proactive robot must be *answerable* at a bad moment, and the reply channel — not the sensing — is where the interaction succeeds or fails.

Proactive human-robot interaction is a well-surveyed field, and the timing and desirability of robot initiative have both received attention [1, 6]. Inference of latent states from physiological signals is likewise mature: heart-rate variability and electrodermal activity are established correlates of stress and arousal [3, 5], and interaction driven by sensed context rather than explicit input has been a design theme since Schmidt's account of implicit interaction [11]. What is less common is a system that carries a latent state all the way from a wristband to a physical drink in a real kitchen, and rarer still is explicit treatment of how the person is meant to *answer* once the robot has decided to speak.

We built BrewBot to find out what that costs. BrewBot is a Kinova Gen3 6-DOF arm hanging from a motorised ceiling rail in `TODO: official lab name`, an instrumented kitchen laboratory at LMU Munich. A heart-rate sensor and an Empatica E4 wristband feed a rule-based state estimator; when the estimate crosses into a state worth acting on, the robot greets the user by name, says what it noticed, and offers a drink. The user can accept by speaking, and a locally hosted language model classifies the reply, including counter-offers such as *"no, but I'd like a coffee instead."* They can also answer with a thumbs-up or thumbs-down to a camera. Or they can raise an open palm, which cuts the robot off mid-sentence and hands the entire dialogue over to the arm: it mimes each drink in turn with its own body and waits for a thumb. The chosen drink is then actually fetched — a glass picked up, filled at the sink or dispensed by a networked coffee machine, and handed over.

The system was built by three students in a course project structured around an intensive on-site practical week, with preparatory work on arm control and kitchen modelling done beforehand and refinement afterwards (106 commits, roughly 4,600 lines of Python on ROS 2 Jazzy [8]). We report it here as a design case study rather than an empirical result: no user study has been run, so our claims are about what building the system forced us to design for. Three patterns recurred strongly enough that we believe they generalise beyond drinks. First, interaction modalities should be *substitutable* rather than merely available in parallel: the open palm does not only stop speech, it re-routes the whole dialogue into embodied mime. Second, deciding *what* to offer and deciding *when* to interrupt are different concerns and belong in different components. Third, every subsystem that is not the drink itself must fail open, because in a real lab something is always down. We also record one scoping decision worth reporting: heart-rate variability is the best-documented stress marker available to us, but deriving it from the wristband's raw pulse waveform turned out to be a signal-processing project in its own right, so we dropped it from the state model and took heart rate from a dedicated sensor instead.

2 Background and Related Work

Proactive robots and the cost of initiative. Koch van den Broek and Moeslund's review of proactive HRI shows a field that agrees proactivity is desirable but defines it inconsistently, ranging from anticipating a user's next subtask to initiating an unrelated conversation [6]. Baraglia et al. found that in a collaborative task users preferred a robot that took initiative over one that waited, but that the benefit depended on the robot acting at points where the human was not already committed to an action [1]. Both results point at the same design pressure that shaped BrewBot: initiative is welcome, but only if it is cheap for the user to dispose of. That framing puts unusual weight on the *decline* path, which in a command-driven system barely exists.

Latent state from physiological signals. Heart rate, heart-rate variability and electrodermal activity are long-standing markers of stress and sympathetic arousal. Healey and Picard classified driver stress from these signals in a real-world setting [3], and Kim et al.'s meta-analysis confirms the HRV–stress relationship while documenting how much it varies across individuals and measurement protocols [5]. That variance matters for a robot: absolute skin conductance differs by roughly two orders of magnitude between people, so any usable rule must be relative to a personal baseline rather than to a population threshold. Validation work on the specific device we used is instructive about where the effort sits. The Empatica E4's electrodermal and temperature channels hold up reasonably in a standardised assessment [13], whereas obtaining trustworthy HRV parameters from its wrist photoplethysmography is a considerably more delicate exercise [12]. That asymmetry determined which signals we ended up using (Section 3.3).

Implicit and multimodal interaction. Schmidt described implicit interaction as a system acting on sensed context that the user did not intend as input [11]; a physiological trigger is a direct instance. Oviatt's critique of multimodal interaction is equally relevant, and in a way we did not expect. Her central point is that users do not blend modalities constantly; they *switch* between them, and they switch precisely when the situation makes one channel expensive [10]. A person holding a glass under a running tap is exactly that situation. This is the observation that BrewBot's answer layer is built around.

Language models in embodied systems. Recent work grounds language models in robot capability so that free-form speech maps onto executable skills [4]. BrewBot uses a far smaller version of the same idea: a locally hosted model performs two jobs, generating the robot's greeting and classifying the user's reply into a closed label set supplied by the caller. We keep the model strictly outside the control path — it never selects a motion — which is what allows the whole dialogue layer to be absent without stopping the drink.

3 The BrewBot System

3.1 Setting and Platform

BrewBot operates in `TODO: official lab name`, a U-shaped instrumented kitchen containing a sink, a networked espresso machine and a workstation. The robot is a Kinova Gen3 6-DOF arm with a Robotiq 2F-140 gripper, mounted upside down on an Elmo carriage that travels along a ceiling rail and can be raised and lowered. Because the arm hangs, its base-frame z axis is inverted relative to intuition, and because it rides the rail, its base frame moves through the room during a task — two facts that shape the whole motion layer.

Sensing is split across three devices. A dedicated Bluetooth Low Energy heart-rate sensor publishes beats per minute. An Empatica E4 wristband provides electrodermal activity, skin temperature and three-axis acceleration, and its act of connecting doubles as an arrival signal. Three cameras run gesture recognition: one on the arm's wrist, one at the workstation, one in the kitchen area. Speech is captured by a microphone and transcribed by Vosk; the robot speaks through Piper. A workstation on the lab network runs Ollama with Llama 3.2, and Home Assistant bridges the espresso machine (over Home Connect) and the ceiling lights.

3.2 Architecture

Figure 2 shows the node graph. Read left to right it is a conventional sense–infer–interact–act pipeline, and each stage is a separate ROS

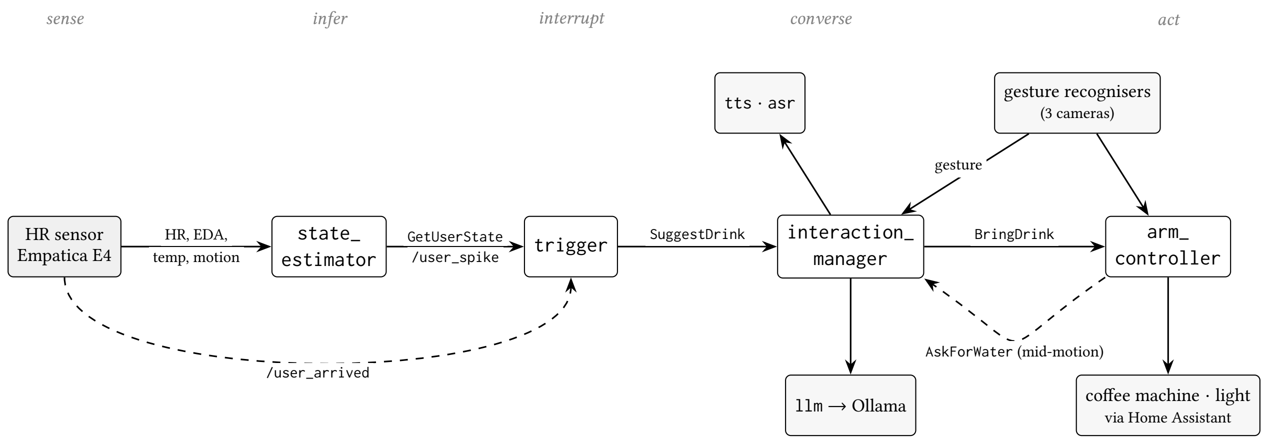

Figure 2: BrewBot's ROS 2 graph. Solid arrows are the forward path from sensing to action; dashed arrows are the two feedback paths that make the system interactive rather than a pipeline — the wristband's arrival event, and the arm calling back into the dialogue layer *while holding a glass under the tap.*

2 node communicating over typed actions and services. Two properties of this decomposition did real work for us and are not visible from the node names.

The first is that `state_estimator` is *passive*. It caches sensor readings and answers a `GetUserState` service; it never sends a goal to anyone. All policy about interrupting a human lives in `trigger`. The second is the dashed arrow returning from `arm_controller` to `interaction_manager`: the arm asks the dialogue layer to talk to the user in the middle of a motion, while the higher-level dialogue goal that started the motion is still open. That single re-entrant edge is what makes the sink interaction possible, and it is why both nodes run multi-threaded executors.

3.3 From Signals to a Suggestion

The estimator reduces four cached signals to a state and a drink using a pure decision function of about twenty lines. Thermal signals are consulted first: skin temperature at or above 35 °C yields *hot* and suggests water, at or below 31 °C yields *cold* and suggests tea. Smoothed motion is consulted next, yielding *active*. Only then is arousal read: a heart rate at least 10 bpm above the user's baseline, or an electrodermal reading at least 15% above its own running baseline, yields *stressed* and suggests tea; an electrodermal reading 15% *below* baseline yields *fatigued* and suggests coffee. Anything else is *calm* and produces no offer at all.

The ordering is the substantive design decision. Heat drives electrodermal activity and movement drives heart rate, so reading arousal first would let *hot*, *active* and *stressed* compete for the same evidence. Excluding the two physical confounds before interpreting arousal is what keeps the rule set honest. The electrodermal thresholds are relative to an exponentially weighted baseline seeded on the first sample, for the between-subject reasons discussed in the literature [5]; dry electrodes also need minutes to equilibrate, which a session-start baseline would capture at exactly the wrong moment.

Two signals named in our original design are absent from that rule set, and the reason is worth stating plainly. The E4 exposes no heart-rate characteristic — only a 64 Hz blood-volume-pulse waveform — so both heart rate and HRV would have had to be derived in our own code. The waveform carries the pulse; the difficulty is not the sensor but the analysis. Recovering beat-to-beat intervals precise enough for HRV from wrist photoplethysmography is a signal-processing task in its own right, and HRV is defined on exactly those intervals. Our first attempt at beat detection illustrates the gap: replaying a reference recording, intervals scattered between 0.51 and 1.22 seconds where the true value sat near 1.03. Autocorrelation over an eight-second window recovered a usable pulse *rate* of 55–64 bpm, but not usable intervals. Doing this properly was a larger undertaking than the rest of the estimator combined, for the estimator's least load-bearing input, so we made the pragmatic call: drop HRV from the state model, and take heart rate from a dedicated BLE sensor that reports beats per minute directly. The wristband kept the channels it measures without inference — electrodermal activity, skin temperature and acceleration — which is also where the published validation is strongest [12, 13].

3.4 Deciding When to Interrupt

`trigger` owns interruption policy and has four ways in. The wristband connecting publishes an arrival event; until that fires, nothing can start a conversation, because there is nobody to talk to. Thereafter the estimator's 1 Hz self-check can publish a spike event, an optional slow poll re-asks every 60 seconds, and a keyboard interface lets an experimenter force any drink.

Spike detection deserves a note because it is a small idea that removed a large class of misbehaviour. The estimator's decision function already encodes "elevated", so the spike detector does not re-implement a threshold; it watches that function's output cross from producing no suggestion to producing one. It fires when three consecutive samples agree — heart rate from a wrist is noisy, and one stray sample must not start a conversation — and only once per episode, with a five-minute cooldown. A user who is simply *still* stressed generates no further offers. Reactive proactivity without

(a) (b) (c)

Figure 3: Modality substitution. (a) The user raises an open palm, cutting the robot off mid-sentence. (b) Rather than abandoning the offer, the arm mimes each drink with its own body. (c) The user answers with a thumbs-up. The conversation continues, having changed modality on the user's terms.

this property degrades into nagging, which is the failure mode the initiative literature warns about [6].

3.5 Speaking and Understanding

The dialogue layer is deliberately small. A single `AskLLM` service carries a prompt and a system prompt, and its behaviour switches on whether the system field is empty. Empty means the language model uses its own barista persona and writes the greeting, at temperature 0.6 so the same sentence is not repeated three times in a row. Filled means the caller has turned the node into a classifier and named the exact labels it will accept, at temperature 0 so a given utterance always classifies the same way.

The language-model node is stateless: no session, no history. Context is assembled by the caller, which sends the pair *"the robot asked X / the person said Y"*. This is not an optimisation but the reason the system can handle a counter-offer. *"No, but I'd like a coffee instead"* is a refusal and an order at once, and only parses correctly when the question is in view; a bare yes/no classifier — which is what we used before, based on sentence embeddings — resolved it to "no" and walked away. The classifier's label set is therefore {YES, NO} plus every drink on the menu, so naming a different drink is itself a valid answer.

Everything in this layer returns the empty string on failure — unreachable model, timeout, or an answer outside the label set. Each caller's decision is binary, use the answer or fall back to a fixed line, and that decision is the same whichever way it failed. The diagnostic reason is logged where it occurred and never crosses an interface boundary. With the language model down, BrewBot still asks its question, in canned wording, and still fetches what it is told.

3.6 Answering in Whatever Modality Is Free

This is the part of BrewBot we consider its contribution. Every question the robot asks goes through one function, which speaks a line, opens the microphone, and simultaneously accepts hand gestures from a shared topic fed by all three cameras. Gestures are recognised with MediaPipe's pre-trained recogniser [7], restricted to thumbs-up, thumbs-down and open palm, and confirmed only after three consecutive frames.

Crucially, the meaning of a gesture is supplied *per question* rather than fixed globally. When the robot offers a drink, an open palm

Figure 4: Waiting at the sink. The arm holds the glass under the tap while the dialogue layer listens for "done". Because a stationary robot reads as a crashed robot, the wait escalates: a wrist wiggle every 15 seconds, then the lab light begins to blink after 40.

means stop; when the robot is holding a glass under a running tap, both an open palm *and* a thumbs-up mean "done", because a person mid-pour will not free a hand to produce a specific one. The same physical gesture answers different questions differently, which is what lets one dialogue primitive serve situations with very different physical constraints.

The open palm is the clearest case (Figure 3). It publishes a stop event that drains the speech queue and terminates audio playback, so the robot falls silent mid-sentence. But it does not cancel the offer. It sets a flag on the fetch request, and the arm then performs the menu physically: for each drink in turn it shapes the gripper, executes a two-pose miming motion, turns to face the user, adds a small wrist tilt that reads as a questioning head-tilt, and waits ten seconds for a thumb. A thumbs-up selects that drink; a thumbs-down moves to the next; a second open palm ends the offer entirely. The estimator's own suggestion is mimed first, because it is the drink the sensors indicate is most likely — not because the user has already heard it named. Frequently they have not: the palm often arrives before the robot reaches the end of its sentence, which is precisely the point of having it.

Which way the arm turns is decided from person detection on the two fixed cameras, running EfficientDet-Lite0 at 2 Hz. The arm faces the kitchen only when the kitchen camera sees someone and the workstation camera does not; the workstation is the default and the tie-break. The wrist camera is deliberately excluded from this decision — what it sees is a fact about where the arm is pointing, not about where the user is standing — but it does contribute to recognition, since all three cameras publish into the same gesture topic. An answer can therefore be seen whether or not the arm is facing the user. The turn is not for perception; it is for legibility. A 6-DOF arm holding still is indistinguishable from a 6-DOF arm that has finished, and the swing plus the small wrist tilt are the only cues that a question is outstanding and the pause belongs to the person.

3.7 Fetching the Drink

Fetching is a sequence of skills over one arm, one gripper and one rail. Named poses are stored in joint space and are executed either through MoveIt [2] for collision-aware planning or through a direct trajectory controller, both driven from the same table so the two backends cannot disagree about the destination. Rail and arm are planned separately: the arm is first moved to a compact tuck pose that is safe by construction, the carriage then travels with the arm frozen, and the kitchen's collision geometry is re-published afterwards because the planning frame itself has moved.

Water is the interesting case, because BrewBot cannot operate the tap. It picks up a glass, travels to the sink, holds the glass under the tap, and asks the person to fill it — which is the re-entrant call in Figure 2. A robot standing still holding a glass looks exactly like a robot that has crashed, so the wait is an escalation ladder (Figure 4): every 15 seconds the wrist performs a small wiggle, after 40 seconds the ceiling light begins to blink, and after roughly two minutes the arm gives up and returns the glass. Both nudges fail open: no light bridge or no trajectory controller costs the feedback, never the water. Coffee and tea take the other route, through a Home Connect espresso machine; tea is the machine's hot-water programme at 70 °C with the bag added by hand. If the machine is unreachable, the arm says so aloud and holds its pose for a fixed interval so a human can pour instead — again, degraded but not stopped.

4 Design Reflections

4.1 Make modalities substitutable, not merely available

Offering speech *and* gestures is not the same as letting one stand in for the other. Oviatt's observation that users switch channels when a channel becomes expensive [10] showed up in our system as a concrete requirement: the reply is often given in the modality the user's body happens to have free, and that is not knowable in advance. Two design commitments followed. Gesture meanings are per-question rather than global, so a thumbs-up can mean "yes" to an offer and "finished" at a tap. And refusing a channel is not refusing the interaction — the palm means "stop talking", so the robot stops talking and asks with its arm instead. Building this cost us one shared gesture topic and one extra field on the fetch request. The concept was cheap; only its consequences for timing were not.

4.2 Separate what to offer from when to interrupt

Our first version had the estimator send goals directly when it detected a state. That conflates two questions with different owners. What the user needs is a function of sensor readings. Whether now is an acceptable moment to interrupt depends on arrival, recency, whether a conversation is already running, and whether the same offer was declined two minutes ago — none of which the estimator can see. Making the estimator a passive service and moving all policy into a trigger node meant the cooldown, the arrival latch and the once-per-episode spike rule were all written once, in one place, and could be changed without touching inference. For a class of system where the entire risk is being annoying, we would now treat this split as the default rather than a refinement.

4.3 Let everything that is not the goal fail open

In a shared lab, on any given day, something is unavailable: the language-model host, the microphone, the coffee machine, the light bridge, the motion planner. A system that treats these as preconditions is a system that mostly does not run. BrewBot therefore degrades along a fixed rule: anything that is not the drink itself fails open. No language model means canned wording. No dialogue stack means the arm hands over an empty glass instead of wedging. No coffee machine means the robot announces the problem and waits for a human. Reviewing the code afterwards, a substantial share of the physical layer is fallback path rather than nominal path, and we consider that proportion appropriate rather than regrettable.

4.4 Keep the language model out of the control path

The language model writes what the robot says and interprets what it hears, and does nothing else. It never selects a motion, and it cannot introduce a drink. The menu lives in a single table that gives, for each drink, the word a person says for it and the route that produces it — the sink, or a named programme on the coffee machine. That one table is simultaneously what the estimator may suggest, what the reply classifier will accept as an answer, what the arm knows how to mime and fetch, and what the machine will brew. Each component verifies itself against it at start-up, so a drink that cannot actually be produced fails at launch rather than halfway through a motion with a glass already in the gripper. The effect is that the least predictable component in the system can only ever affect wording and interpretation, both of which have defined fallbacks. Given how readily a small model answers slightly outside the labels it was given — and it does, however firmly the prompt forbids it — we would not have been comfortable with a looser boundary.

5 Limitations

BrewBot has not been evaluated with users. Everything above is a design account supported by development experience, not by measurement, and the patterns we propose are hypotheses about what matters rather than findings. The estimator's thresholds are principled but uncalibrated: they were chosen from the literature and from resting values, not fitted against self-reported ground truth, and there is no per-user calibration period. The drink poses are hard-coded in joint space, so the world must not move — a glass placed 10 cm from where it belongs is not picked up. Sessions were demonstration-length rather than deployments, and a single trigger policy was in use throughout, so we cannot say how the offer rate would feel over a working day. Finally, physiological data is sensitive personal data; our prototype logged it locally in a research setting, and any longer-term deployment would need a consent and retention story we did not have to build.

6 Future Work

The single highest-leverage next step is fiducial-based perception. Placing AprilTags [9] in the kitchen and on the drink station would let a one-shot calibration anchor the fixed cameras and the objects in a room frame, which the arm can then convert into its own moving base frame from the rail's position feedback. This one addition unlocks three capabilities we currently lack, and they compound. Hand-over and drop-off become dynamic: the robot can deliver where the cup actually is, rather than to a fixed pose the user must walk to. Cup pick-up becomes autonomous rather than dependent on a glass being pre-placed within a millimetre budget. And, most interesting for the interaction, the arm can look into the cup and judge whether it is empty. That last one closes a loop the physiological layer cannot close on its own: finishing a drink is a far better refill trigger than any wrist signal, and it converts BrewBot from a system that offers once into one that keeps a person supplied.

Beyond perception, three directions follow from the limitations. A short per-user calibration period would replace population thresholds with personal ones, which the between-subject variance in the literature suggests is where most of the accuracy is [5]. Breadth is the other axis: every additional drink currently needs a validated motion sequence and, where the machine is involved, a confirmed programme identifier, and a per-user preference table would let the same physical state map to different drinks for different people — tea for one person's fatigue, coffee for another's. And the obvious study is now designable: a within-subjects comparison of proactive against command-driven service in the same kitchen, measuring perceived intrusiveness, trust and task load, with a second factor varying which reply modalities are offered. Our claim that substitutability matters more than availability is exactly the kind of claim such a study would test.

7 Conclusion

BrewBot infers a latent physical state from wearable sensors and decides on its own that a person might want a drink — but it then *asks*. It says what it noticed, names the drink it has in mind, and fetches something only once the person has accepted, whether they accept in words, with a thumb, or by picking a different drink from the menu the arm mimes at them. The offer, not the delivery, is the interaction. Building it end to end moved our attention from the sensing question we started with to a design question we did not anticipate: once a robot speaks first, the interaction lives or dies on how cheaply the person can answer, in whatever modality their hands and attention happen to allow. Treating modalities as substitutable, separating what to offer from when to interrupt, and letting every non-essential subsystem fail open were the three commitments that made an unsolicited robot tolerable to work next to. We offer them as design guidance for state-aware service robots, and the open questions they raise as an agenda for evaluating one.

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